

Measure Theory and Ergodic Theory

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The foundational knowledge relating to Measure & Ergodic Theory, that has not been covered elsewhere, was provided by Robertson (2023).

Definition 0.1 (Gleason, 2010, sect. 2). A *topological measure space* $(X, \mathcal{B}, \mu)$ is a topological space (X, τ) such that $\mathcal{B}$ is generated by the open sets defined by the topology τ , i.e., $\mathcal{B} = \text{Borel}(X) = \sigma(\tau)$, and μ is a measure on this space.

A *Borel measure* is the measure μ on a topological measure space $(X, \mathcal{B}, \mu)$ where (X, τ) is Hausdorff.

A *regular (Borel) measure* is a measure on a Borel measure space $(X, \mathcal{B}, \mu)$ such that the following hold:

1. *Finite Compact Measure*: For any compact subset $K \subseteq X$, then $\mu(K) < \infty$.
2. *Outer Regularity*: For any $B \in \mathcal{B}$, then

$$\mu(B) = \inf\{\mu(C) \mid B \subseteq C, C \text{ is open}\}.$$

3. *Inner Regularity*: For any $U \in \tau$, or, in other words, any open subset $U \subseteq X$, then

$$\mu(U) = \sup\{\mu(K) \mid K \subseteq U, K \text{ compact}\}.$$

A *left-Haar measure* [or *right-Haar measure*] on a topological group (Γ, τ) is a non-zero regular Borel measure μ on Γ such that $\mu(\gamma \cdot B) = \mu(B)$ [or $\mu(B \cdot \gamma) = \mu(B)$] for all $\gamma \in \Gamma$ and $B \in \sigma(\tau)$.

1 Eigenfunctions

Let $(X, \text{Bor}(X), \mu, T)$ be a measure-preserving system.

Definition 1.1 (Kra et al., 2026, sect. 2). A nonzero function $f \in L^2(X, \text{Bor}(X), \mu)$ is a *measurable eigenfunction* with eigenvalue $\theta \in \mathbb{C}$ if $f(Tx) = \theta f(x)$ holds for μ -almost every $x \in X$.

Definition 1.2 (Kra et al., 2026, sect. 2). A nonzero function $f \in \mathcal{C}(X)$ is a *topological eigenfunction* with eigenvalue $\theta \in \mathbb{C}$ if $f(Tx) = \theta f(x)$ for all $x \in X$.

2 Discrete Spectrum

Proposition 2.1 (Jamneshan and Kreidler, 2025, Corollary 5.1.11). *Let $U : \Gamma \rightarrow \mathcal{U}(H)$ be a unitary representation of an abelian group Γ . Then*

$$H_{ds} = \bar{\text{lin}} \bigcup_{\chi \in \Gamma^*} \ker(\chi - U).$$

3 Mixing

Definition 3.1 (Kra et al., 2026, sect. 2). We define $\mathcal{K}$ to be the sub- σ -algebra of $\text{Bor}(X)$ where $L^2(X, \mathcal{K}, \mu)$ is the closed subspace of $L^2(X, \text{Bor}(X), \mu)$ spanned by its measurable eigenfunctions.

Proposition 3.1 (cf. Kra et al., 2026, sect. 2). *For any $E \in \text{Bor}(X)$ and μ -almost every $x \in E$, we have $\mathbb{E}(1_E \mid \mathcal{K})(x) > 0$.*

Theorem 3.1 (Kra et al., 2026, Theorem 2.4). *For any $f \in L^2(X, \text{Bor}(X), \mu)$, we have*

$$f \text{ is weak mixing} \iff \mathbb{E}(f \mid \mathcal{K}) = 0.$$

Gleason, J. (2010). *Existence and uniqueness of haar measure*.

Jamneshan, A. and Kreidler, H. (2025). *ISem 28: Ergodic structure theory and applications*.

Kra, B. et al. (2026). *A short proof of erdős's $B + C$ conjecture*.

Robertson, D. (2023). *MATH41021/61021 measure and ergodic theory*. University of Manchester. Available at: <https://personalpages.manchester.ac.uk/staff/donald.robertson/teaching/23-24/41021>